

1. There is no statement of a problem. So, it is not clear the purpose of the experiment.
2. I think any reader would feel there is too much going on. The abstract talks about abilities, overestimation of correlations, social class. It is a lot.

Ideally, in the abstract you follow a organized structure:

State a problem > say what we do > give a key result > talk about implications.

Peer skill identification and social class: Evidence from a referral field experiment*

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January 9, 2025

Abstract

Using performance-pay incentives in a field experiment with 849 university students, we study how well peers identify cognitive and social skills of their classmates at the end of the semester. We find that students can identify cognitive skills and struggle with social skills. They overestimate the correlation between skills and use academic performance as a proxy to refine their guesses. In the treatment condition, we provide additional incentives for identifying high-skilled peers from lower social classes. We find limited evidence for a social class bias in nominations, and the treatment moderately increases nominations for lower social class individuals for the biased referrers. This effect appears driven by students' limited ability to accurately identify peers' social class.

JEL Classification:

Keywords: behavioral economics, referrals, field experiment, skills, inequality

*We obtained Institutional Review Board approvals from NYU Abu Dhabi (HRPP 2024-50) and the University of Luxembourg (ERP 24-028). The study design was preregistered in the OSF Registries prior to data collection (see <https://doi.org/10.17605/OSF.IO/V9T3W>).

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1 Introduction

Referrals are an important channel for connecting workers with firms, transmitting valuable skill information to form good matches. Approximately half of all jobs are filled through referrals (Topa, 2011), and referred workers tend to earn higher wages, have longer tenure, and are more likely to be hired than non-referred candidates.¹ Referral hiring can also propagate inequality due to its inherent reliance on networks (Bolte, Immorlica, & Jackson, 2024; Jackson, 2021), with lower social classes especially at the risk of being left out (Chetty et al., 2022b; Stansbury & Rodriguez, 2024).

Yet the precise mechanisms driving how and when referrals provide positive information about worker quality are still not well understood. Skilled workers may know other skilled workers, and refer them as a consequence of their network composition (Montgomery, 1991). Alternatively, given the network composition, certain workers may be better at screening productivity, with screening ability depending on the skill in question. Existing empirical work focuses on naturalistic settings with endogenous network formation, making it difficult to disentangle screening ability from network composition effects (e.g., Beaman and Magruder (2012); Pallais and Sands (2016)). In this paper, we focus on the latter and ask two questions: Can individuals identify productive peers at different skills when network compositions are held constant? Second, because referrals are prone to biases (Beaman, Kelcher, & Magruder, 2018), do disadvantaged individuals from lower social classes still face referral barriers when network and performance differences are accounted for?

We conducted a lab-in-the-field experiment in a private Colombian university to answer these questions. Participating bachelor students were asked to refer classmates they interacted with throughout the semester in small classrooms (average 26 students per class). This particular setting allowed us to fix network compositions and made referral sets fully observable to researchers. We first collected incentivized cog-

¹See Topa (2019) for a review of the literature. Examples include Bayer, Ross, and Topa (2008); Brown, Setren, and Topa (2016); Dustmann, Glitz, Schönberg, and Brücker (2016); Friebe, Heinz, Hoffman, and Zubanov (2023); Hellerstein, Kutzbach, and Neumark (2014, 2019); Schmutte (2015).

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Author: Manu Munoz Date: 09/01/2025, 16:54:37
this statement does not follow from your previous paragraph. you introduced biases and now you are talking about worker quality.

Author: Manu Munoz Date: 09/01/2025, 16:54:34

Author: Manu Munoz Date: 09/01/2025, 16:56:01

Author: Manu Munoz Date: 09/01/2025, 16:56:07
this also does not follow from the previous statement.

when you say previous research has a limitation. Your immediate statement should be how to solve it. Or how you solve it.

Author: Manu Munoz Date: 09/01/2025, 16:57:53
Talking about different skills in the research question is a bit odd because you have not had a discussion or a mention on different skills being something important.

Maybe streamline the first paragraphs to allow your research questions to follow from each thing you have said.

Also, I am not sure the key thing we offer is a constant network structure. think that each element and word in your research question should be linked to the problem, the contribution, the design, etc.

Author: Manu Munoz Date: 09/01/2025, 16:57:51

Author: Manu Munoz Date: 09/01/2025, 16:57:19

Author: Manu Munoz Date: 09/01/2025, 17:00:05
What does this mean? accounting for network and performance differences?

Maybe think about the research questions from a wide angle. I think that you start with two good points in referrals and that is part of why we did what we did. You say: (i) referrals help matching - so they are good to identify quality, (ii) referrals can be biased - so they are bad at promoting equity/diversity.

With this in mind, I would go straight to connecting to it, and saying: it is important to know if they work across multiple dimensions of quality, for example, for different types of skills (although ideally then we could start by saying that the labor market is more complex now and different types of skills are needed, including cognitive and social, among others).

Then, I would say, well if these skills are important and we want to evaluate how good are referrals in identifying them, we would like to see that also through the lens of equity. So, you can point towards the need of understanding if the existing biases towards certain groups are widespread or skill specific.

Then, your research questions could be about that: How good are people in making referrals for social and cognitive skills? and how likely are referrers to display biases in their selection - focus specifically on social class?

Author: Manu Munoz Date: 09/01/2025, 17:06:16
Try to use first person. When you say observable to researchers, it feels you were displaying these networks to other researchers different to us.

Author: Manu Munoz Date: 09/01/2025, 17:06:13

utive and social skill measures, and then asked for private nominations of the referral candidate. ~~Referred students did not take any action, and with incentives contingent on referral performance we ruled out concerns of potential social transfers (i.e., nepotism or favoritism) or reputational costs.~~ Once we abstracted away from these elements, the referral exercise resembled to one of measuring productivity beliefs through nominated candidates where biases against lower social classes could still be at play. ~~We introduced quota-like incentives as a treatment arm to increase referrals and mitigate these potential biases.~~

We find that referrers can screen for cognitive but not for social skill. ~~Our empirical strategy consists of normalizing the number of referrers across classrooms and comparing the shares of referrals a participant receives in each skill.~~ Students with higher scores on the cognitive skill measure get more referrals, but we find no evidence that social skill can be screened effectively in the same way. ~~Research hiring may help identifying some skills but not others depending on the context.~~ Crucially, we find that a large majority of referrers screen for academic performance instead: Students with better grades get a substantially higher share of referrals when controlling for either skill. Participants have differential referral strategies that either focus on screening for GPA only, screening for GPA and cognitive skill, or neither of the two. The ability to screen for grades in our sample is better than the ability to screen for skills. ~~Our findings highlight that referrals do convey positive information, but the type of information being conveyed is assessed and may not necessarily match what the firm requires.~~

The support for a bias affecting lower social classes is limited. Only the group of referrers screening for cognitive skill and GPA refer a lower share of their low-SES classmates when productivity and network compositions are accounted for. ~~The remaining majority of referrals are not biased, despite our experimental setting in a highly unequal society with significant class inequalities and limited class mobility (Angulo, Gaviria, Pérez, & Azevedo, 2012; García, Rodríguez, Sánchez, & Bedoya, 2015).~~ The treatment partially mitigates the bias for this subset of referrers, while not changing the referral rates of lower social classes in the rest. Heterogeneous ability to identify social class may

Author: Manu Munoz Date: 09/01/2025, 17:07:08

Maybe break it. This sentence explains the two main elements of our experimental sessions. Say we measure performance on each skill and explain why. Then say, once we have a full map of performance for all referrers and potential referees, we asked participants to make referrals...

Author: Manu Munoz Date: 09/01/2025, 17:08:27

Author: Manu Munoz Date: 09/01/2025, 17:08:40

Again, you are putting too much information on our key contribution - the design, in one single sentence.

If you're gonna talk about those referred not doing anything, explain it. say that the measurement allowed us to have performance for everyone before the referral, and why that is useful.

Tjhen say that we incentivized referrals based on the performance of those referred, and why.

Author: Manu Munoz Date: 09/01/2025, 17:08:38

Author: Manu Munoz Date: 09/01/2025, 17:10:45

Author: Manu Munoz Date: 09/01/2025, 17:10:46

This is also super important. Do not cramp it into a single sentence. Treatments are your design. So, you need to explain it. Say we run two treatments, a control... a treatment/quota... each intending to...

Author: Manu Munoz Date: 09/01/2025, 17:11:45

Author: Manu Munoz Date: 09/01/2025, 17:11:48

Not sure these details are for an introduction

Author: Manu Munoz Date: 09/01/2025, 17:12:54

Author: Manu Munoz Date: 09/01/2025, 17:12:56

What does this mean? As we do not have hiring in our experiment, you should first state the results clearly and cleanly.

Then you can offer some discussion if you want.

Author: Manu Munoz Date: 09/01/2025, 17:14:07

You are using the word screen/screening many many times.

I also don't know how informative it is. The way you are presenting the results you are telling the reader what the participants were thinking/looking for. But we do not know that exactly. A better way to present results is to say what we find... so, instead of saying participant screened on GPA, you could say something like, we find a strong effect of GPA in predicting who gets a referral (or something like that).

Author: Manu Munoz Date: 09/01/2025, 17:14:05

Author: Manu Munoz Date: 09/01/2025, 17:18:20

Author: Manu Munoz Date: 09/01/2025, 17:16:38

Again, not sure how this connects to firms.

Maybe keep it more focused on skills, ability to refer, biases, and the things you said we would do. In the discussion of the paper at the end you could conjecture something about firms if you think it can be connected.

Author: Manu Munoz Date: 09/01/2025, 17:18:24

Prioritize active instead of passive voice.

We do not find support for a social class bias... or something like that. And then explain why, in a way that people can understand without reading the paper. Currently the statement of productivity and network composition are not too clear. Because they have not been defined... at least composition hasn't

Author: Manu Munoz Date: 09/01/2025, 17:20:21

Author: Manu Munoz Date: 09/01/2025, 17:20:25

This is not that evident. The society may be very unequal and still students at the university may not be. We are not running an experiment with general population and a representative sample. We have a specific group of students from a private university in Bucaramanga.

Author: Manu Munoz Date: 09/01/2025, 17:21:33

Comments from page 3 continued on next page

cognitive and social skill measures, and then asked for private nominations of the referral candidate. Referred students did not take any action, and with incentives contingent on referral performance we ruled out concerns of potential social transfers (i.e., nepotism or favoritism) or reputational costs. Once we abstracted away from these elements, the referral exercise resembled to one of measuring productivity beliefs through nominated candidates where biases against lower social classes could still be at play. We introduced quota-like incentives as a treatment arm to increase referrals and mitigate these potential biases.

We find that referrers can screen for cognitive but not for social skill. Our empirical strategy consists of normalizing the number of referrers across classrooms and comparing the shares of referrals a participant receives in each skill. Students with higher scores on the cognitive skill measure get more referrals, but we find no evidence that social skill can be screened effectively in the same way. Referral hiring may help identifying some skills but not others depending on the context. Crucially, we find that a large majority of referrers screen for academic performance instead: Students with better grades get a substantially higher share of referrals when controlling for either skill. Participants have differential referral strategies that either focus on screening for GPA only, screening for GPA and cognitive skill, or neither of the two. The ability to screen for grades in our sample is better than the ability to screen for skills. Our findings highlight that referrals do convey positive information, but the type of information being conveyed is nuanced and may not necessarily match what the firm requires.

The support for a bias affecting lower social classes is limited. Only the group of referrers screening for cognitive skill and GPA refer a lower share of their low-SES classmates when productivity and network compositions are accounted for. The remaining majority of referrals are not biased, despite our experimental setting in a highly unequal society with significant class inequalities and limited class mobility (Angulo, Gaviria, Pérez, & Azevedo, 2012; García, Rodríguez, Sánchez, & Bedoya, 2015). The treatment partially mitigates the bias for this subset of referrers, while not changing the referral rates of lower social classes in the rest. Heterogeneous ability to identify social class may

play a role in the efficiency of the treatment. **Our findings show referrals are robust to salient differences between social classes, and provide evidence that existing biases can be remedied with changes in the referral incentive structure.**

We contribute to the broader literature on referral experiments, notably about the productivity information transmitted by referrals. Past work provides causal evidence that referrals have higher productivity compared to non-referred workers (Pallais & Sands, 2016), especially under incentives contingent on referral performance that reduce social transfers between the referrer and referrals (Beaman & Magruder, 2012; Witte, 2021). Our simplified setting eliminates social transfers as well as reputational concerns, and highlights the skill-dependent screening ability of referrers under performance contingent incentives with fixed networks. **This specific aspect of referral ability to screen for skills has not been studied in isolation.**

We also contribute to the growing body of work on the relevance of noncognitive skills in the labor market. Patience, self-control, conscientiousness, teamwork, and critical thinking are skills that contribute to positive labor market returns (Heckman & Kautz, 2012; Heckman, Stixrud, & Urzua, 2006; Lindqvist & Vestman, 2011; Weinberger, 2014). Interpersonal skills among them are exceptionally relevant for labor market gains in the last two decades as a complement to cognitive skill (Deming, 2017, 2023). Hiring firms report consistent difficulties in observing social skills among candidates and applicants are willing to pay substantial sums to convey social skill feedback to employers (Bassi & Nansamba, 2022). While referrers can screen for cognitive skill and job-specific abilities, but not perseverance (Beaman et al., 2018; Beaman & Magruder, 2012; Pallais & Sands, 2016), evidence on whether referrals can help screening for social skills is missing. In our setting, **we examine whether referrals can facilitate screening for both cognitive and social skills in a highly educated sample.**

Our results speak to the literature on the diversity considerations in the referral literature. Network homophily² in referrals drives correlations among social groups'

²Homophily is a well-documented empirical consistency in sociology where individuals form ties more often with others who are similar to themselves across observable characteristics (McPherson, Smith-

Author: Manu Munoz Date: 09/01/2025, 17:22:17

Author: Manu Munoz Date: 09/01/2025, 17:22:19
I thought the results did not point to a bias towards social class. So, this is surprising to me.

Author: Manu Munoz Date: 09/01/2025, 17:25:29

Author: Manu Munoz Date: 09/01/2025, 17:23:33
This is our key contribution, not the transfers and the fixed networks.

Make this clear earlier. State how studying referrals for different abilities in the same setting: we ask the same referrer to nominate for different abilities using the same set of potential nominees is novel and valuable. But more importantly, contributions are not based on the design, or at least not mainly, contributions to the literature are on the findings. So here, you should also state what we find relevant to this literature.

Author: Manu Munoz Date: 09/01/2025, 17:25:31
something related to this should be stated above, at the beginning, to motivate our study... I think I left you a comment above about it.

Author: Manu Munoz Date: 09/01/2025, 17:26:20

Author: Manu Munoz Date: 09/01/2025, 17:26:21
Again, the contribution must be linked to findings...

employment and wages (Calvo-Armengol & Jackson, 2004; Calvo-Armengol & Jackson, 2007). Individuals are tied more often to others with comparable socioeconomic status (Chetty et al., 2022b). Limited interaction across social classes due to spatial segregation is shown to drive at least some of the differences in outcomes (Chetty et al., 2022a), but we consider a setting where networks across social classes overlap by design. Anticipating differences in referral outcomes for low-SES even when spatial constraints are removed and productivity is accounted for, we introduce quota-like incentives as a treatment arm to increase referrals to low-SES. We design the treatment incentives in inspiration from the success of gender quotas in the affirmative action literature (e.g., Balafoutas and Sutter (2012); Bertrand, Black, Jensen, and Lleras-Muney (2019); Niederle, Segal, and Vesterlund (2013)).

Section 2 begins with the background and setting in Colombia. In Section 3 we present the design of the experiment, including the skill assessment, referral and guessing tasks. In Section 4 we describe the data and procedures. Section 5 discusses the results of the experiment. Section 6 concludes. The Appendix presents additional tables and figures as well as the experiment instructions.

2 Background and Setting

Colombia has consistently ranked as one of the most unequal countries in Latin America (World Bank, 2024), with the richest decile earning 50 times more than the poorest decile (United Nations, 2023). This economic disparity is reflected by a highly stratified society, where social classes remain separated by education and geographic residence.³ While imperfect -because it disregards the income of residents, the official stratification system largely aligns with and potentially reinforces existing social class divisions (Guevara S Lovin, & Brashears, 2006; McPherson, Smith-Lovin, & Cook, 2001).

³In 1994, Colombia introduced a nationwide classification system dividing the population into 6 strata based on housing characteristics and neighborhood amenities. This system, initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), now extends to university fees and social program eligibility (stratum 4 neither receives subsidies nor pays extra taxes).